ABSTRACT

Student dropout is a major challenge in the education system, affecting both individuals and society. Early identification of students at risk can help institutions take preventive measures and improve academic outcomes. This project aims to develop a machine learning-based system to predict student dropout risk using academic and socio-economic data.The dataset used includes various attributes such as study time, absences, parental education, and student performance. A new target variable called “dropout” is created based on the final grade. Data preprocessing techniques such as cleaning and encoding are applied to prepare the dataset for analysis.Exploratory Data Analysis (EDA) and visualization techniques are used to identify patterns and relationships between variables. A Logistic Regression model is implemented to classify students into at-risk and not at-risk categories. The model is evaluated using performance metrics such as accuracy, precision, recall, and confusion matrix.The results show that factors like study time, absences, and academic performance significantly influence dropout risk. This project demonstrates how machine learning can be effectively used to support early intervention strategies and improve student retention.This work aligns with Sustainable Development Goal 4 (Quality Education) by promoting data-driven decision-making in the education sector.